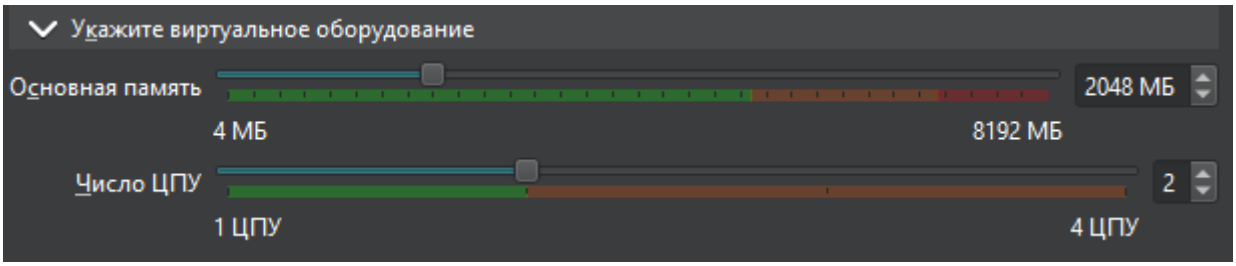
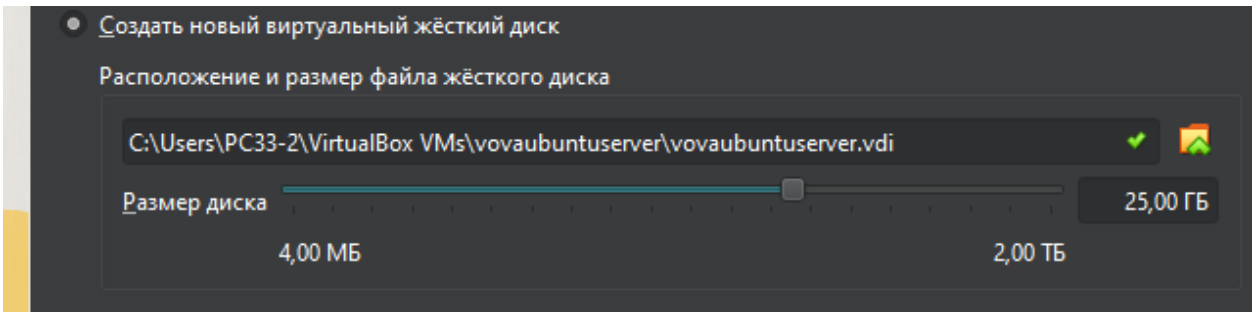
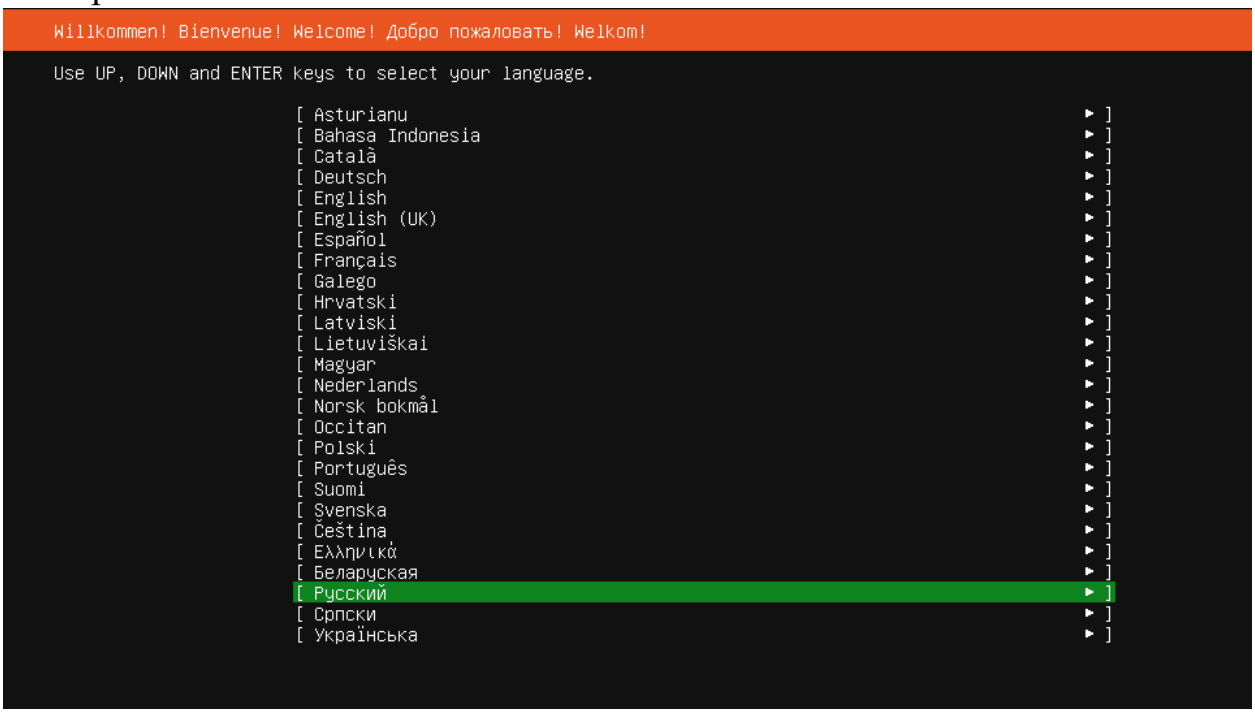
День 6

1. Установка ubuntu server:





Выбор языка:



Логин пароль:

```
Profile configuration                                                    [ Help ]

Enter the username and password you will use to log in to the system. You can configure SSH access on a later screen, but a
password is still needed for sudo.

                Ваше имя:  vova

        Your servers name:  vovaserver
                            The name it uses when it talks to other computers.

Введите имя пользователя:  vovak

           Задайте пароль:  *

       Подтвердите пароль:  *_
```

Установка системы:

```
Установка системы

 curtin command in-target
installing system
 executing curtin install initial step
 executing curtin install partitioning step
  curtin command install
   configuring storage
    running 'curtin block-meta simple'
     curtin command block-meta
      removing previous storage devices
      configuring disk: disk-sda
     configuring partition: partition-0
     configuring partition: partition-1
     configuring format: format-0
     configuring partition: partition-2
     configuring lvm_volgroup: lvm_volgroup-0
     configuring lvm_partition: lvm_partition-0
     configuring format: format-1
     configuring mount: mount-1
     configuring mount: mount-0
 executing curtin install extract step
  curtin command install
   writing install sources to disk
    running 'curtin extract'
     curtin command extract
      acquiring and extracting image from cp:///tmp/tmp3gduvrqz/mount
 configuring keyboard
  curtin command in-target
 executing curtin install curthooks step
  curtin command install
   configuring installed system
    running 'curtin curthooks'
     curtin command curthooks
      configuring apt configuring apt
      installing missing packages
      configuring iscsi service
      configuring raid (mdadm) service
      configuring NVMe over TCP
      installing kernel
      setting up swap
      apply networking config
      writing etc/fstab
      configuring multipath \
```

Вход в систему:

```
Ubuntu 25.04 vovaserver tty1

vovaserver login: vovak
Password:
Welcome to Ubuntu 25.04 (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Вт 28 апр 2026 06:04:54 UTC

  System load:             0.03
  Usage of /:              44.1% of 11.21GB
  Memory usage:            13%
  Swap usage:              0%
  Processes:               119
  Users logged in:         0
  IPv4 address for enp0s3: 10.0.2.15
  IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe74:d180

66 обновлений может быть применено немедленно.
Чтобы просмотреть дополнительные обновления выполните: apt list --upgradable



The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

vovak@vovaserver:~$
```

2. Установка БД (Redis):

Обновление пакетов:

```
vovak@vovaserver:~$ sudo apt update
[sudo] password for vovak:
Сущ:1 http://ru.archive.ubuntu.com/ubuntu plucky InRelease
Сущ:2 http://ru.archive.ubuntu.com/ubuntu plucky-updates InRelease
Сущ:3 http://ru.archive.ubuntu.com/ubuntu plucky-backports InRelease
Сущ:4 http://security.ubuntu.com/ubuntu plucky-security InRelease
Пол:5 http://ru.archive.ubuntu.com/ubuntu plucky/main Translation-en [519 kB]
Пол:6 http://ru.archive.ubuntu.com/ubuntu plucky/main Translation-ru [358 kB]
Пол:7 http://ru.archive.ubuntu.com/ubuntu plucky/restricted Translation-ru [72
Пол:8 http://ru.archive.ubuntu.com/ubuntu plucky/restricted Translation-en [13
Пол:9 http://ru.archive.ubuntu.com/ubuntu plucky/universe Translation-en [6 28
Пол:10 http://ru.archive.ubuntu.com/ubuntu plucky/universe Translation-ru [1 4
Пол:11 http://ru.archive.ubuntu.com/ubuntu plucky/multiverse Translation-ru [6
Пол:12 http://ru.archive.ubuntu.com/ubuntu plucky/multiverse Translation-en [1
Пол:13 http://ru.archive.ubuntu.com/ubuntu plucky-updates/main Translation-en
Пол:14 http://ru.archive.ubuntu.com/ubuntu plucky-updates/restricted Translati
Пол:15 http://ru.archive.ubuntu.com/ubuntu plucky-updates/universe Translation
Пол:16 http://ru.archive.ubuntu.com/ubuntu plucky-updates/multiverse Translati
Пол:17 http://ru.archive.ubuntu.com/ubuntu plucky-backports/universe Translati
Получено 9 058 kB за 2с (4 065 kB/s)
Может быть обновлено 66 пакетов. Запустите «apt list --upgradable» для их пока
```

Установка:

```
Для их удаления используйте «sudo apt autoremove».

Установка:
  redis-server

Установка зависимостей:
  liblzf1  redis-tools

Предлагаемые пакеты:
  ruby-redis

Сводка:
  Обновление: 0, Установка: 3, Удаление: 0, Пропуск обновления: 66
  Объём загрузки: 1 384 kB
  Требуемое пространство: 7 584 kB / 6 024 MB доступно

Пол:1 http://ru.archive.ubuntu.com/ubuntu plucky/universe amd64 liblzf1 amd64 3
Пол:2 http://ru.archive.ubuntu.com/ubuntu plucky-updates/universe amd64 redis-
Пол:3 http://ru.archive.ubuntu.com/ubuntu plucky-updates/universe amd64 redis-
Получено 1 384 kB за 0с (3 214 kB/s)
Выбор ранее не выбранного пакета liblzf1:amd64.
(Чтение базы данных … на данный момент установлено 126686 файлов и каталогов.)
Подготовка к распаковке …/liblzf1_3.6-4_amd64.deb …
Распаковывается liblzf1:amd64 (3.6-4) …
Выбор ранее не выбранного пакета redis-tools.
Подготовка к распаковке …/redis-tools_5%3a7.0.15-3ubuntu0.1_amd64.deb …
Распаковывается redis-tools (5:7.0.15-3ubuntu0.1) …
Выбор ранее не выбранного пакета redis-server.
Подготовка к распаковке …/redis-server_5%3a7.0.15-3ubuntu0.1_amd64.deb …
```

Проверка статуса сервера:

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
vovak@vovaserver:~$ sudo systemctl status redis-server
● redis-server.service - Advanced key-value store
     Loaded: loaded (/usr/lib/systemd/system/redis-server.service; enabled; pr
     Active: active (running) since Tue 2026-04-28 06:09:58 UTC; 3min 9s ago
 Invocation: 972988f1165f44baa0ed23825205b9be
       Docs: http://redis.io/documentation,
             man:redis-server(1)
   Main PID: 1711 (redis-server)
     Status: "Ready to accept connections"
      Tasks: 5 (limit: 1873)
     Memory: 4.2M (peak: 4.7M)
        CPU: 332ms
     CGroup: /system.slice/redis-server.service
             └─1711 "/usr/bin/redis-server 127.0.0.1:6379"

апр 28 06:09:58 vovaserver systemd[1]: Starting redis-server.service - Advance
апр 28 06:09:58 vovaserver systemd[1]: Started redis-server.service - Advanced
vovak@vovaserver:~$
```

3. Подключение к БД (локально):

```
апр 28 06:39:36 vovaserver systemd[1]: Start
апр 28 06:39:37 vovaserver systemd[1]: Start
vovak@vovaserver:~$ redis-cli -a "1"
Warning: Using a password with '-a' or '-u'
127.0.0.1:6379> PING
PONG
127.0.0.1:6379> _
```

Подключение к БД (удалённо(Через PuTTy):

```
vovak@vovaserver: ~
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.3.51' (ED25519) to the list of known hosts.
vovak@192.168.3.51's password:
Welcome to Ubuntu 25.04 (GNU/Linux 6.14.0-37-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Вт 28 апр 2026 07:22:41 UTC

  System load:  0.0                Processes:               126
  Usage of /:   45.4% of 11.21GB   Users logged in:          0
  Memory usage: 15%                IPv4 address for enp0s3: 192.168.3.51
  Swap usage:   0%


65 обновлений может быть применено немедленно.
Чтобы просмотреть дополнительные обновления выполните: apt list --upgradable


Last login: Tue Apr 28 07:21:32 2026 from 192.168.3.204
vovak@vovaserver:~$
```

4. Импортировать данные в БД:

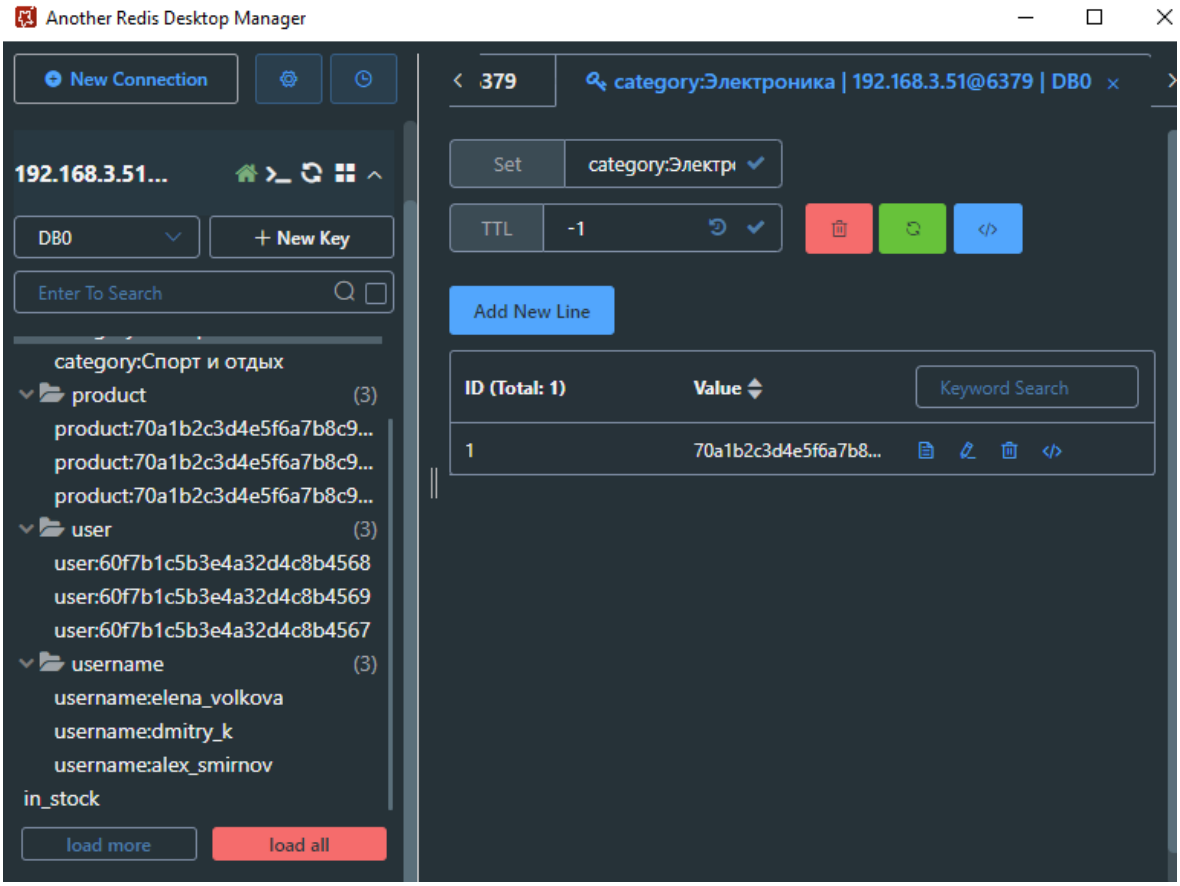
```
Last login: Tue Apr 28 07:21:32 2026 from 192.168.3.204
vovak@vovaserver:~$ redis-cli -a "1" << EOF
> HSET product:70a1b2c3d4e5f6a7b8c9d000 name "Механическая клавиатура Logitech M
X" category "Электроника" price 12990 in_stock 1 warehouse_qty 45 specs '{"color
":"черный","switch_type":"Brown","backlight":"RGB"}' tags "клавиатура,работа,под
арок"
> HSET product:70a1b2c3d4e5f6a7b8c9d001 name "Термокружка Stanley 0.5L" category
 "Спорт и отдых" price 3490 in_stock 1 warehouse_qty 120 specs '{"color":"зелены
й","volume_ml":500,"material":"нержавеющая сталь"}' tags "термокружка,поход,авто
"
> HSET product:70a1b2c3d4e5f6a7b8c9d002 name "Книга 'Чистый код' Р. Мартин" cate
gory "Книги" price 890 in_stock 0 warehouse_qty 0 specs '{"pages":464,"language"
:"русский","cover":"мягкая"}' tags "книга,программирование,обучение"
>
```

Проверка того что все данные загрузились:

```
vovak@vovaserver:~$ redis-cli -a "1"
Warning: Using a password with '-a' or '-u' option on the command line interface
 may not be safe.
127.0.0.1:6379> KEYS product:*
1) "product:70a1b2c3d4e5f6a7b8c9d002"
2) "product:70a1b2c3d4e5f6a7b8c9d001"
3) "product:70a1b2c3d4e5f6a7b8c9d000"
127.0.0.1:6379>
```

```
127.0.0.1:6379> KEYS user:*
1) "user:60f7b1c5b3e4a32d4c8b4567"
2) "user:60f7b1c5b3e4a32d4c8b4568"
3) "user:60f7b1c5b3e4a32d4c8b4569"
127.0.0.1:6379>
```

```
127.0.0.1:6379> KEYS product:*
1) "product:70a1b2c3d4e5f6a7b8c9d002"
2) "product:70a1b2c3d4e5f6a7b8c9d001"
3) "product:70a1b2c3d4e5f6a7b8c9d000"
127.0.0.1:6379> HGETALL product:70a1b2c3d4e5f6a7b8c9d000
 1) "name"
 2) "\xd0\x9c\xd0\xb5\xd1\x85\xd0\xb0\xd0\xbd\xd0\xb8\xd1\x87\xd0\xb5\xd1\x81\xd
0\xba\xd0\xb0\xd1\x8f \xd0\xba\xd0\xbb\xd0\xb0\xd0\xb2\xd0\xb8\xd0\xb0\xd1\x82\x
d1\x83\xd1\x80\xd0\xb0 Logitech MX"
 3) "category"
 4) "\xd0\xad\xd0\xbb\xd0\xb5\xd0\xba\xd1\x82\xd1\x80\xd0\xbe\xd0\xbd\xd0\xb8\xd
0\xba\xd0\xb0"
 5) "price"
 6) "12990"
 7) "in_stock"
 8) "1"
 9) "warehouse_qty"
10) "45"
11) "specs"
12) "{\"color\":\"\xd1\x87\xd0\xb5\xd1\x80\xd0\xbd\xd1\x8b\xd0\xb9\",\"switch_ty
pe\":\"Brown\",\"backlight\":\"RGB\"}"
13) "tags"
14) "\xd0\xba\xd0\xbb\xd0\xb0\xd0\xb2\xd0\xb8\xd0\xb0\xd1\x82\xd1\x83\xd1\x80\xd
0\xb0,\xd1\x80\xd0\xb0\xd0\xb1\xd0\xbe\xd1\x82\xd0\xb0,\xd0\xbf\xd0\xbe\xd0\xb4\
```


Another Redis Desktop Manager
New Connection
379
category:Электроника | 192.168.3.51@6379 | DB0
192.168.3.51...
DB0
+ New Key
Enter To Search
category:Спорт и отдых
product (3)
product:70a1b2c3d4e5f6a7b8c9...
product:70a1b2c3d4e5f6a7b8c9...
product:70a1b2c3d4e5f6a7b8c9...
user (3)
user:60f7b1c5b3e4a32d4c8b4568
user:60f7b1c5b3e4a32d4c8b4569
user:60f7b1c5b3e4a32d4c8b4567
username (3)
username:elena_volkova
username:dmitry_k
username:alex_smirnov
in_stock
load more
load all
Set
category:Электр
TTL
-1
Add New Line
ID (Total: 1)
Value
Keyword Search
1
70a1b2c3d4e5f6a7b8...

5. Установка API сервисов (Fusio):

Развертывание Fusio:

```
vovak@vovaserver:~/fusio$ sudo docker compose up -d
[+] up 0/65
 ⠼ Image m...
[+] up 0/65sio/fusio Pulling                                          2.0s
 ⠼ Image m...
[+] up 0/65s
 ⠼ Image m...
[+] up 0/65s
 ⠼ Image m...
[+] up 0/65s
 ⠼ Image m...
[+] up 0/65s
 ⠼ Image m...
[+] up 0/65s
 ⠼ Image m...
[+] up 0/65s
 ⠼ Image m...
[+] up 0/65s
 ⠼ Image m...
[+] up 0/65s
 ⠼ Image m...
[+] up 4/65s
 ⠼ Image m...
[+] up 10/65
 ⠼ Image m...
[+] up 15/65
 ⠼ Image m...
[+] up 18/65
 ⠼ Image m...
[+] up 25/65
 ⠼ Image m...
[+] up 29/65
 ⠼ Image m...
[+] up 36/65
 ⠼ Image m...
[+] up 40/65
 ⠼ Image m...
[+] up 40/76
 ⠼ Image m... [           ]
[+] up 40/76
 ⠼ Image m... [           ]
[+] up 40/76
 ⠼ Image m... [           ]
[+] up 40/76
 ⠼ Image m... [           ]
[+] up 40/76
 ⠼ Image m... [           ]
[+] up 40/76
 ⠼ Image m... [           ]
[+] up 40/76
 ⠼ Image m... [           ]
[+] up 40/76
 ⠼ Image m... [           ]
[+] up 40/76
 ⠼ Image m... [           ]
```

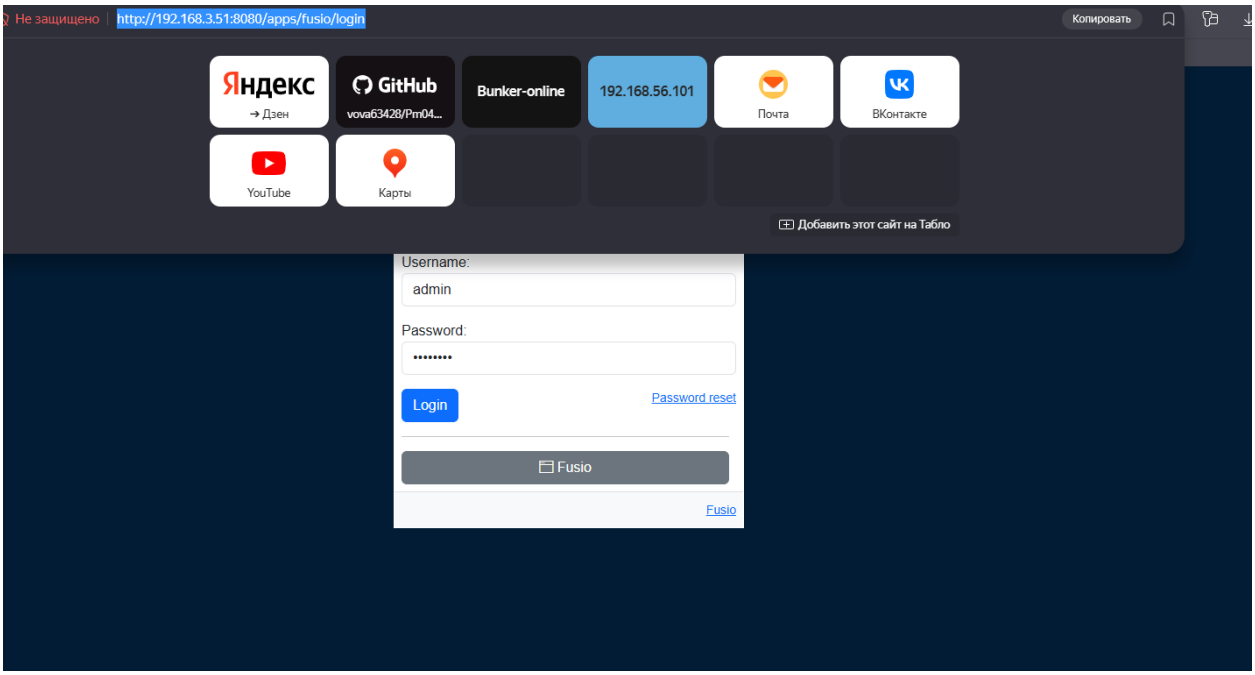
Скачать и запустить:

```
vovak@vovaserver:~$ git clone https://github.com/mendableai/firecrawl
Cloning into 'firecrawl'...
remote: Enumerating objects: 76262, done.
remote: Counting objects: 100% (733/733), done.
remote: Compressing objects: 100% (282/282), done.
remote: Total 76262 (delta 590), reused 451 (delta 450), pack-reused 75529 (from 2)
Receiving objects: 100% (76262/76262), 85.59 MiB | 10.80 MiB/s, done.
Resolving deltas: 100% (58374/58374), done.
vovak@vovaserver:~$ cd firecrawl
vovak@vovaserver:~/firecrawl$ cp apps/api/.env.example .env
vovak@vovaserver:~/firecrawl$ # Обязательно отключи аутентификацию
vovak@vovaserver:~/firecrawl$ sed -i 's/USE_DB_AUTHENTICATION=true/USE_DB_AUTHENTICATION=false/' .env
vovak@vovaserver:~/firecrawl$ # Запускаем всё одной командой
vovak@vovaserver:~/firecrawl$ sudo docker compose up -d
sudo: docker: command not found
```

Данные:

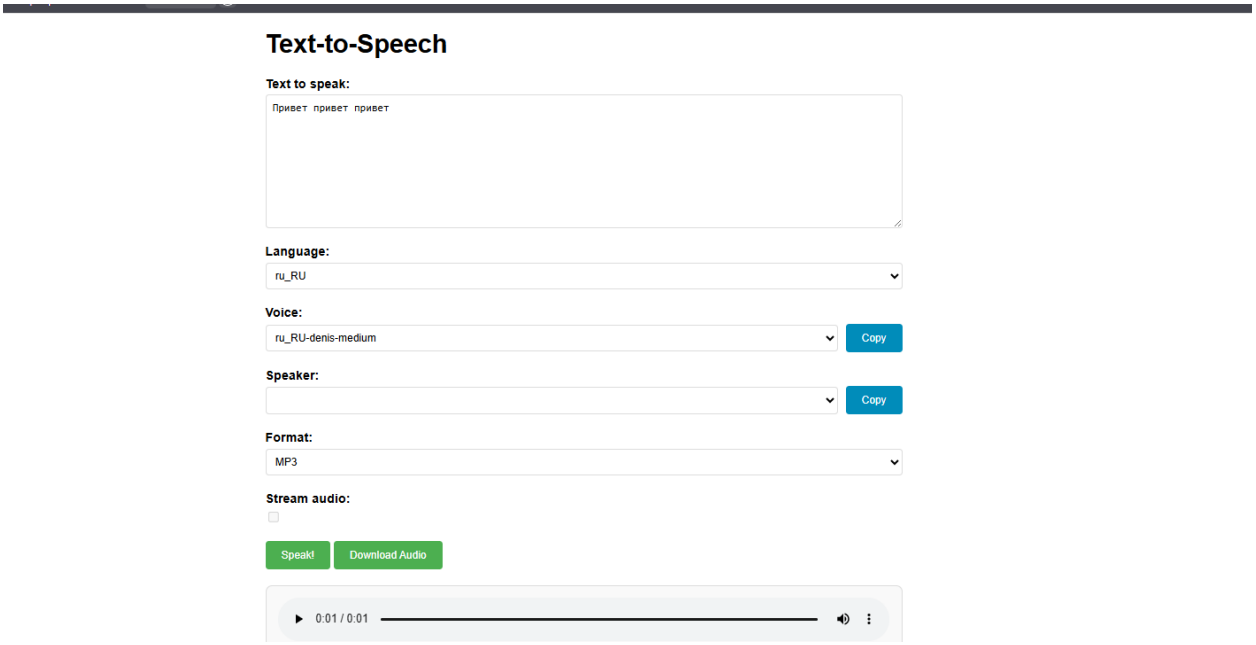
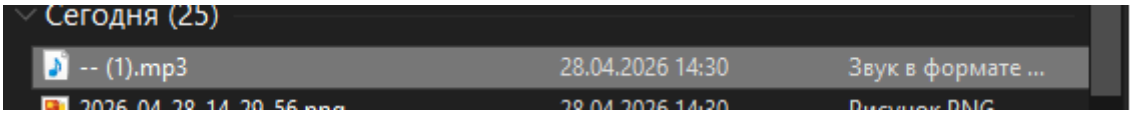
```
  GNU nano 8.3
services:
  fusio:
    image: fusio/fusio
    restart: always
    environment:
      FUSIO_PROJECT_KEY: "42eec18ffdbffc9fda6110dcc705d6ce"
      FUSIO_CONNECTION: "pdo-mysql://fusio:fusio123@mysql/fusio"
      FUSIO_BACKEND_USER: "vova"
      FUSIO_BACKEND_EMAIL: "admin@localhost.com"
      FUSIO_BACKEND_PW: "1"
    ports:
      - "8080:80"
    depends_on:
      - mysql
    networks:
      - fusio-net

  mysql:
    image: mysql:8.0
    restart: always
    environment:
      MYSQL_RANDOM_ROOT_PASSWORD: "1"
      MYSQL_USER: "fusio"
      MYSQL_PASSWORD: "fusio123"
      MYSQL_DATABASE: "fusio"
    volumes:
      - ./db:/var/lib/mysql
    networks:
      - fusio-net

networks:
  fusio-net:
```

Вводим в браузер:




```
{
  "apiVersion": "8.7.4.0",
  "title": "Fusio",
  "description": "Self-Hosted API Management for Builders.",
  "paymentCurrency": "EUR",
  "categories": [
    "authorization",
    "backend",
    "consumer",
    "default",
    "system"
  ],
  "scopes": [
    "default"
  ],
  "links": [
    {
      "rel": "root",
      "href": "http:\/\/192.168.3.51:8080\/"
    },
    {
      "rel": "apps",
      "href": "http:\/\/192.168.3.51:8080\/apps"
    },
    {
      "rel": "openapi",
      "href": "http:\/\/192.168.3.51:8080\/system\/generator\/spec-openapi"
    },
    {
      "rel": "typeapi",
      "href": "http:\/\/192.168.3.51:8080\/system\/generator\/spec-typeapi"
    },
    {
```

```
        "rel": "route",
        "href": "http:\/\/192.168.3.51:8080\/system\/route"
      },
      {
        "rel": "health",
        "href": "http:\/\/192.168.3.51:8080\/system\/health"
      },
      {
        "rel": "oauth2",
        "href": "http:\/\/192.168.3.51:8080\/authorization\/token"
      },
      {
        "rel": "whoami",
        "href": "http:\/\/192.168.3.51:8080\/authorization\/whoami"
      },
      {
        "rel": "api-catalog",
        "href": "http:\/\/192.168.3.51:8080\/.well-known\/api-catalog"
      },
      {
        "rel": "oauth-authorization-server",
        "href": "http:\/\/192.168.3.51:8080\/.well-known\/oauth-authorization-server"
      },
      {
        "rel": "oauth-protected-resource",
        "href": "http:\/\/192.168.3.51:8080\/.well-known\/oauth-protected-resource"
      },
      {
        "rel": "openid-configuration",
        "href": "http:\/\/192.168.3.51:8080\/.well-known\/openid-configuration"
      },
      {
        "rel": "security",
        "href": "http:\/\/192.168.3.51:8080\/.well-known\/security.txt"
      },
      {
        "rel": "about",
        "href": "https:\/\/www.fusio-project.org"
      }
    ]
}
```

6. TTS AI + API:

```
vovak@vovaserver:~/piper-tts$ sudo docker ps | grep piper
vovak@vovaserver:~/piper-tts$ sudo docker logs piper-server
vovak@vovaserver:~/piper-tts$ curl -X POST http://localhost:8080/api/tts \
>   -H "Content-Type: application/json" \
>   -d '{"text": "Привет! Это тест русского голоса.", "voice": "ru_RU-irinia-low
"}' \
>   --output test.wav
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100   198  100    97  100   101   5961   6207 --:--:-- --:--:-- --:--:-- 12375
vovak@vovaserver:~/piper-tts$ sudo docker run -d \
>   --name piper-server \
>   -p 8081:8080 \
>   -v "$(pwd)/voices:/voices" \
>   --restart unless-stopped \
>   nbr23/gopipertts
docker: Error response from daemon: Conflict. The container name "/piper-server"
 is already in use by container "39dd14c90fe3f42218159071201704a5a1b7e55e49cc313
7e21b2e39f290f11b". You have to remove (or rename) that container to be able to
reuse that name.

Run 'docker run --help' for more information
vovak@vovaserver:~/piper-tts$ ^C
vovak@vovaserver:~/piper-tts$ sudo docker rm -f piper-server
piper-server
vovak@vovaserver:~/piper-tts$ sudo docker run -d \
>   --name piper-server \
>   -p 8081:8080 \
>   -v "$(pwd)/voices:/voices" \
>   --restart unless-stopped \
>   nbr23/gopipertts
c88ac40d649311e3b1b5301b63161c535b145a163d6ac4d4ea08c36c3405e21e
vovak@vovaserver:~/piper-tts$ sudo docker ps | grep piper
c88ac40d6493   nbr23/gopipertts   "/app/gopipertts"        19 seconds ago   Up 1
9 seconds (health: starting)   0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp   pip
er-server
vovak@vovaserver:~/piper-tts$
```